Sustainability & recycling

National 5 · Machines, materials & finishing · Worksheet

WOODWORK WITH MR KELLY

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ explain the benefits of sustainably sourced timber
- ☐ describe ways to make a product more environmentally friendly
- ☐ follow workshop recycling practices

I DO — Worked example

Read how a full-mark answer is built

A manufacturer makes children's garden chairs.

1 Describe two things the manufacturer could do to make the chair more environmentally friendly. (2 marks)

Full-mark answer each ✓ is one mark

Use FSC-certified wood from forests that are replanted ✓. Plan the cutting list so that waste is reduced, and use offcuts for small parts ✓.

Why marks are not awarded

"Make it greener and recycle it."

0 marks: too vague.

WE DO — Guided practice

Use the starters and word bank to help you

A shop sells shelves made from FSC-certified pine.

2 Describe a benefit of using sustainably sourced wood. (1 mark)

Sentence starters

One … is …, which means …

Another … is …, so …

Word bank

forests · replanted · wildlife · maintained · habitats · supply · protected · long-term

Tip: 1 marks = 1 separate, developed points.

YOU DO **On your own** No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) Which would make a product more environmentally friendly?

☐ Planning the cutting list to reduce waste ☐ Using a solvent-based varnish finish ☐ Using an imported tropical hardwood ☐ Using extra glue in every joint

b) What is an environmental advantage of chipboard?

☐ It's waterproof ☐ It's made from waste wood ☐ It has attractive grain ☐ It contains no glue

c) How should you dispose of leftover solvent finish?

☐ Down the sink with plenty of water ☐ In the general waste bin ☐ Safely, following workshop procedures ☐ Poured onto the ground outside

d) What does the FSC logo show?

☐ The wood is waterproof and needs no finish ☐ The wood is a hardwood, not a softwood ☐ The wood is the cheapest available ☐ The wood comes from responsibly managed forests

4 Key terms: write the letter of the matching description next to each term.

1. **FSC**	____	**A** Trees are replanted and habitats protected
2. **Benefit of sustainable timber**	____	**B** Contains resins, harder to recycle
3. **Reducing waste**	____	**C** Forest Stewardship Council: wood from responsibly managed forests
4. **Eco-friendly finish**	____	**D** Made from waste wood
5. **Chipboard environmental advantage**	____	**E** Water-based finishes give off fewer fumes
6. **Chipboard environmental disadvantage**	____	**F** Plan the cutting list and use offcuts

A cabinet could be made from chipboard or pine.

5 Describe one environmental advantage and one disadvantage of chipboard compared with pine. (2 marks)

A workshop produces a lot of offcuts and sawdust.

6 Describe two ways the workshop could reduce or recycle this waste. (2 marks)

Check and reflect

Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?